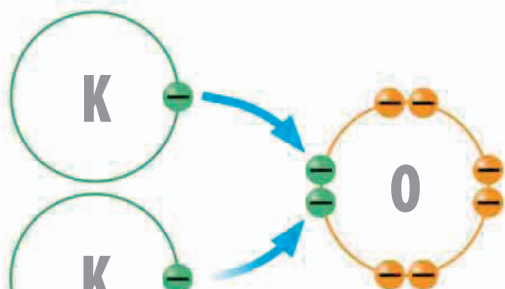


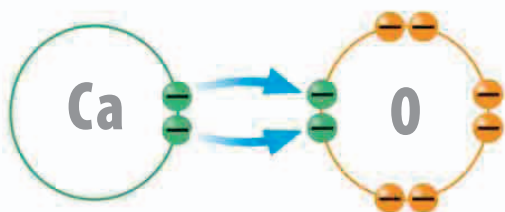
Balancing charges

For a bond to form between ions, the positive and negative charges need to balance so that the overall molecule is neutral. As a result compounds are not always derived from one anion and one cation, but form with different proportions of ions.



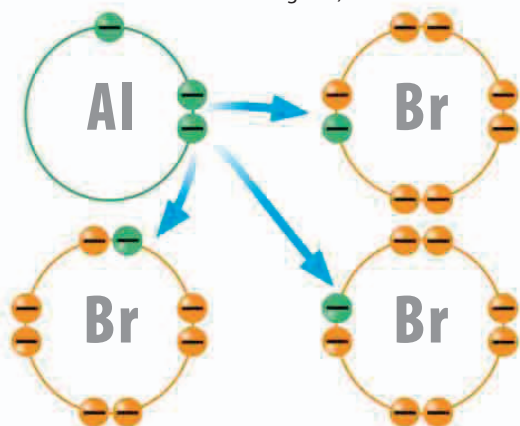
△ Potassium oxide

Two potassium ions ($2 \times \text{K}^+$; total charge $2+$) bond to one oxygen ion ($1 \times \text{O}^{2-}$; total charge $2-$), to form a neutral molecule.



△ Calcium oxide

One calcium ion (Ca^{2+} ; total charge $2+$) bonds to one oxygen ion (O^{2-} ; total charge $2-$).



△ Aluminum bromide

One aluminum ion ($1 \times \text{Al}^{3+}$; total charge $3+$) bonds to three bromine ions ($3 \times \text{Br}^-$; total charge $3-$).



Reactivity

Metal atoms give away electrons so they are electropositive. Nonmetals gain electrons so they are electronegative. Different atoms give away or gain electrons more easily than others.

▽ Metal ions

Magnesium (Mg) and sodium (Na) have three electron shells. However, magnesium has two outer electrons while sodium has one. It takes less energy to lose one electron than two, so sodium is more electropositive than magnesium. Potassium (K) also has one outer electron but it is in a fourth shell, farther away from the attractive pull of the nucleus. So potassium loses its outer electron more easily than sodium.

▽ Nonmetal ions

Oxygen (O) needs two electrons to complete the octet but nitrogen (N) needs three. It takes less energy to gain two electrons than three, so oxygen is more electronegative than nitrogen. Phosphorus (P) also needs three electrons but it has one more shell. The pull from the nucleus in this third shell is weaker than in a second shell, so it is harder for phosphorus to gain electrons than nitrogen.

